

Yen Giang Nguyen

Van Andel Institute | Department of Neurodegenerative Science
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EDUCATION

B.A., Biology & Mathematics, *magna cum laude*

2023 - 2026

Kalamazoo College, Kalamazoo, MI

- **GPA:** 3.842 | Dean's List
- **Concentration:** Neuroscience & Biological Physics
- **Coursework:** Linear Algebra, Multivariable Calculus, Mathematical Statistics, Topology & Real Analysis

RESEARCH EXPERIENCE

Research Data Technician

Jul 2026 - Now

Henderson Laboratory at Van Andel Institute, Grand Rapids, MI

- Computational analysis and modeling of neurodegenerative disease pathology and progression.

Bioinformatics Intern

Jun 2024 - Jun 2026

Henderson Laboratory at Van Andel Institute, Grand Rapids, MI

❖ **Project 1: CosMx vs. Xenium Comparative Analysis**

- Compared the performance of the 1000-plex CosMx Mouse Neuro Panel to the 247-plex Xenium Mouse Neuro using R on high-performance computing (HPC) cluster.
- Annotated cell types in both datasets using supervised cell typing with the InSituType package, presenting results through flightpath plots, UMAP, and heatmaps.
- Conducted differential expression analysis of CosMx and Xenium through Seurat and MAST packages and visualized comparative results through volcano and violin plots.

❖ **Project 2: VAI Brain Bank Analysis**

- Analyzed quantitative pathology data from ~140 postmortem human brains in the VAI Brain Bank, focusing on hyperphosphorylated tau, amyloid beta (A β), α -synuclein, and TDP-43 accumulation across multiple brain regions using Python.
- Developed a pseudo-longitudinal staging framework using dose-response curve modeling to align patients along disease progression trajectories.
- Performed dimensionality reduction and disease subtyping through principal component analysis (PCA), identifying distinct pathological groupings and patterns of neurodegenerative progression.
- Applied the Subtype and Stage Inference (SuStain) framework to model heterogeneous disease subtypes and progression stages from cross-sectional neuropathology data.

HONORS AND AWARDS

- **Undergraduate Travel Grant, Computational and Systems Neuroscience** 2026
Competitive registration, travel, and lodging support for undergraduate attendance at COSYNE.
- **The Clarke Benedict Williams Prize, Kalamazoo College** 2026
Highest graduating class distinction for achievement in mathematics and allied sciences.
- **Senior Integrated Project Honors, Department of Biology, Kalamazoo College** 2025
Awarded to selected outstanding senior research projects in biology.
- **McColl Endowment Travel Award, Department of Biology, Kalamazoo College** 2025
- **Department of Biology Travel Award, Kalamazoo College** 2024

Travel support to attend Neuroscience 2024.

- **Lux Esto Merit Scholarship, Kalamazoo College** 2023
Four-year merit scholarship awarded for academic achievement.
- **CCIR Spotlight Scholar, Cambridge Centre for International Research** 2022
Selected for distinction in an independently completed research project.

PRESENTATIONS

❖ Oral Presentation

- **Quantitative Brain Banking Offers New Insights into Neurodegenerative Disease Progression**
Nguyen YG, Cook J, Noe C, Abe K, Turner L, Johansson A, Rohrer D, Williams E, Moore D, Flanagan M, Henderson MX. *Grand Challenges in Parkinson's Disease*. 2026

❖ Poster Presentation

- **Quantitative Brain Banking Offers New Insights into Neurodegenerative Disease Progression**
Nguyen YG, Cook J, Noe C, Abe K, Turner L, Johansson A, Rohrer D, Williams E, Moore D, Flanagan M, Henderson MX. *Grand Challenges in Parkinson's Disease*. 2026
- **Comparing Single-Cell Spatial Transcriptomics Techniques**
Nguyen YG, Goralski T, Breton L, Meyerdirk L, Adams M, Henderson MX. *Aligning Science Across Parkinson's - Spatial Transcriptomics Meeting*. 2024
- **Gene Therapy in the Central Nervous System: ImageJ as a Tool for Image Analysis**
Nguyen YG. *Neuroscience 2024, Society for Neuroscience*. 2024

TEACHING EXPERIENCE

- Teaching Assistant** Mar - Jun 2024
Kalamazoo College, Kalamazoo, MI
- Graded and answered questions related to the assignments for the Applied Statistics course.

ADDITIONAL TRAINING

- Neuromatch student in Computational Neuroscience course** 2025
- Completed an intensive three-week, live program focused on practical applications of computational modeling and machine learning in neuroscience.

TECHNICAL SKILLS

Programming: Python, R, LaTeX

Analysis: statistical modeling, principal component analysis, clustering, nonlinear curve fitting, differential-expression analysis, spatial transcriptomics

Libraries: pandas, NumPy, Matplotlib, SuStaln, Seurat, MAST, InSituType, ggplot2

Tools: Git, GitHub, Azure